

DS4420 Final Project Bayesian Model

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Objective

We created a Bayesian predictive model that predicts the likelihood of pet adoption based on the animal's identified key traits.

Read Data

```
pets <- read.csv("pet_adoption_data.csv")
head(pets)
```

```
##   PetID PetType      Breed AgeMonths  Color  Size  WeightKg Vaccinated
## 1   500   Bird    Parakeet     131 Orange  Large  5.039768         1
## 2   501  Rabbit      Rabbit      73  White  Large 16.086727         0
## 3   502   Dog Golden Retriever    136 Orange Medium  2.076286         0
## 4   503   Bird    Parakeet      97  White  Small  3.339423         0
## 5   504  Rabbit      Rabbit     123  Gray  Large 20.498100         0
## 6   505   Dog    Labrador      70  Brown  Large 20.986261         0
##   HealthCondition TimeInShelterDays AdoptionFee PreviousOwner
## 1                0                27          140             0
## 2                0                 8          235             0
## 3                0                85          385             0
## 4                0                61          217             1
## 5                0                28           14             1
## 6                0                87          301             1
##   AdoptionLikelihood
## 1                    0
## 2                    0
## 3                    0
## 4                    0
## 5                    0
## 6                    0
```

Preprocess

```
pets_scaled <- pets

# numeric features get scaled
pets_scaled$AgeMonths <- as.numeric(scale(pets$AgeMonths))
pets_scaled$AdoptionFee <- as.numeric(scale(pets$AdoptionFee))
pets_scaled$WeightKg <- as.numeric(scale(pets$WeightKg))
pets_scaled$TimeInShelterDays <- as.numeric(scale(pets$TimeInShelterDays))

# categorical features
pets_scaled$Breed <- as.factor(pets$Breed)
pets_scaled$PetType <- as.factor(pets$PetType)
pets_scaled$HealthCondition <- as.factor(pets$HealthCondition)
pets_scaled$Vaccinated <- as.factor(pets$Vaccinated)
pets_scaled$PreviousOwner <- as.factor(pets$PreviousOwner)

# sanity check for Bernoulli outcome
stopifnot(all(pets_scaled$AdoptionLikelihood %in% c(0, 1)))
```

Fit Bayes Model

```
library(brms)

## Warning: package 'brms' was built under R version 4.3.3

## Loading required package: Rcpp

## Warning: package 'Rcpp' was built under R version 4.3.3

## Loading 'brms' package (version 2.22.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').

##
## Attaching package: 'brms'

## The following object is masked from 'package:stats':
##
##      ar

model <- brm(
  AdoptionLikelihood ~ AgeMonths + Breed + PetType + HealthCondition + Vaccinated +
    AdoptionFee + TimeInShelterDays + PreviousOwner + WeightKg,
  family = bernoulli("logit"),
  data = pets_scaled,
  chains = 4,
  cores = 4,
```

```

seed = 42,
warmup = 1000,
iter = 2000,
refresh = 100,
control = list(adapt_delta = 0.95, max_treedepth = 12)
)

## Compiling Stan program...

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.1 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Trying to compile a simple C file

## Running "C:/PROGRA~1/R/R-43~1.1/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 12.3.0'
## gcc -I"C:/PROGRA~1/R/R-43~1.1/include" -DNDEBUG -I"C:/Users/jjams/AppData/Local/R/win-library/4.3
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/Core:19
##                  from C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/Dense:1
##                  from C:/Users/jjams/AppData/Local/R/win-library/4.3/StanHeaders/include/stan/math/p
##                  from <command-line>:
## C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10
## 679 | #include <cmath>
##      |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-43~1.1/etc/x64/Makeconf:265: foo.o] Error 1

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.1 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Start sampling

## Warning: There were 3998 transitions after warmup that exceeded the maximum treedepth. Increase max_
## https://mc-stan.org/misc/warnings.html#maximum-treedepth-exceeded

## Warning: Examine the pairs() plot to diagnose sampling problems

## Warning: The largest R-hat is 3.4, indicating chains have not mixed.
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#r-hat

## Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#bulk-ess

## Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quant
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#tail-ess

```

Outcomes

lol uhh,, are we supposed to do anything else here...

```
summary(model)
```

```
## Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
## careful when analysing the results! We recommend running more iterations and/or
## setting stronger priors.
```

```
## Family: bernoulli
## Links: mu = logit
## Formula: AdoptionLikelihood ~ AgeMonths + Breed + PetType + HealthCondition + Vaccinated + AdoptionFee
## Data: pets_scaled (Number of observations: 2007)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error  l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      1399.03    449.51    687.55  2298.83 1.58      7      40
## AgeMonths       -0.58     0.06    -0.71   -0.49 1.10     38     64
## BreedLabrador    2.58     0.33     1.92    3.14 1.43      8     46
## BreedParakeet   -1401.42   449.49  -2301.10 -689.85 1.58      7     40
## BreedPersian    -646.12  1609.80  -3888.17 1714.23 3.40      4     11
## BreedPoodle      0.29     0.27    -0.21    0.81 1.28     11     30
## BreedRabbit     2925.20  3954.65  -4818.02 8600.19 2.73      5     12
## BreedSiamese    -645.90  1609.80  -3888.19 1714.20 3.40      4     11
## PetTypeCat      -755.45  1688.11  -3495.02 2137.59 3.10      5     15
## PetTypeDog     -1401.47   449.50  -2301.10 -689.65 1.58      7     40
## PetTypeRabbit   -4326.79  3927.94 -10339.62 3375.61 2.71      5     15
## HealthCondition1 -2.19     0.18    -2.52   -1.87 1.16     17     29
## Vaccinated1      2.21     0.15     1.92    2.48 1.27     12     36
## AdoptionFee     -0.08     0.07    -0.20    0.02 1.52      7     63
## TimeInShelterDays 0.06     0.06    -0.06    0.17 1.15     19     55
## PreviousOwner1   0.02     0.13    -0.23    0.28 1.53      8     22
## WeightKg        -0.01     0.06    -0.11    0.13 1.13     21     20
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```